OS Unit 3 - Paging & Virtual Memory

Paging: physical memory is divided into fixed-size frames and logical memory into pages of the same size. The page table maps each page number to a frame number, so a process need not be stored contiguously. No external fragmentation; small internal fragmentation in the last page.
Address translation: logical address = (page number p, offset d). TLB caches recent page-table entries.

Virtual memory and demand paging: pages are loaded only when needed. A reference to a page not in memory causes a page fault.

Page replacement algorithms
- FIFO: replace the oldest page. Can show Belady's anomaly (more frames, more faults).
- Optimal page replacement: replace the page that will not be used for the longest time. Lowest fault rate, but needs the future - used as a benchmark.
- LRU (least recently used): replace the page that has not been used for the longest time. Good approximation of optimal; implemented with counters or a stack.

Thrashing: the process spends more time paging than executing because it does not have enough frames. Cause: degree of multiprogramming too high. Fix: working-set model, page-fault frequency control.